

# LESSON 8.3

## SOLVING EQUATIONS WITH VARIABLES ON BOTH SIDES



### LESSON INTRODUCTION

**MA.8.AR.2.1** Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

### LEARNING TARGETS

- I can solve equations with variables on both sides.

### BENCHMARK COHERENCE

#### BUILDING ON

**MA.7.AR.2.2** Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

#### WORKING TOWARD

**MA.912.AR.1.2** Rearrange equations or formulas to isolate a quantity of interest.

**MA.912.AR.2.1** Given a real-world context, write and solve one-variable, multi-step linear equations.

### ESSENTIAL QUESTIONS

- How do you solve an equation with variables on both sides?
- How can you use algebra tiles to model solving equations?

### LESSON NARRATIVE

This lesson builds on students' previous knowledge of solving multi-step equations in one variable. Students begin by exploring how to solve equations with variables on both sides of the equation using algebra tiles. Students will then learn how to solve such equations algebraically before practicing solving equations with variables on both sides on their own. This lesson focuses on mathematical contexts while providing an opportunity for students to model concretely before moving to abstract models.

### COMPONENT SUMMARY

#### 8.3.1 WARM-UP (~5 MIN.)

Use hanger diagrams to model equations

#### 8.3.2 EXPLORATION (~15-20 MIN.)

Use algebra tiles to explore solving equations with variables on both sides

#### 8.3.3 GUIDED INSTRUCTION (10-15 MIN.)

Learn how to solve equations with variables on both sides algebraically

#### 8.3.4 YOUR TURN! (10 MIN.)

Practice solving equations with variables on both sides

#### 8.3.5 WRAP-UP (~5 MIN.)

Reflect on the lesson learning target

## RESOURCES

### GLOSSARY

Key Terms: N/A

Additional Terms: N/A

### MATERIALS

- (Optional) Algebra tiles
- (Optional) Highlighters
- (Optional) Colored pencils

### ADDITIONAL PREPARATION

- If you allow students to use algebra tiles to model solving the equations, have the tiles ready for students to use in pairs.

## TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may incorrectly identify and try to combine terms that can be combined.
- Students may think there is only one order to perform the steps in an equation.
- Students may get stuck when trying to determine where to get started when solving equations with variables on both sides of the equals sign.

### THINGS TO CONSIDER

- Provide opportunities for students to examine multiple approaches to solving the same problem side-by-side. For example, consider modeling more than one approach in 8.3.3 Guided Instruction.

### LITERACY AND LANGUAGE ROUTINES

#### Compare and Connect

To foster students' meta-awareness of different approaches to solving equations, teachers should demonstrate thinking out loud as they model solving an equation in more than one way. Students should be prompted to reflect and respond as they discuss different approaches to solving equations.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

#### MA.K12.MTR.4.1 Discussion and Reflection

Students will be working to solve multi-step equations by using a series of logical steps. Clarifications within this standard emphasize analyzing mathematical thinking and explaining methods and procedures. Encourage students to meta-cognitively reflect on the processes they use and to describe them using correct mathematical vocabulary.

## DIFFERENTIATE INSTRUCTION

Working with algebra tiles allows students to physically manipulate parts of an equation as they work toward solving for the variable. Encourage students to verbally describe the steps they use to solve equations and to discuss the process of solving for an unknown variable.

### 8.3.1 WARM-UP: BALANCING HANGERS

#### TEACHER MOVES

The equations students write for question 1b can and will vary based on whether students combine values or not. It is not the expectation that all students write the same equations, but that they consider equations where variables are on both sides.

Sample responses:

A:  
 $3+6+6+s+6 = s+s+s+s+6+s$   
 $15+s+6=4s+6+s$   
 $21+s=5s+6$

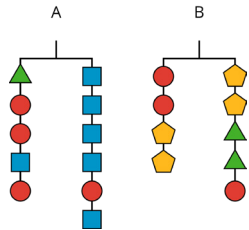
B:  
 $6+6+p+p = p+p+3+3+6$   
 $12+2p=2p+12$

#### QUESTIONING

- Consider asking the following as extension questions:
- Why is it not possible to determine the weight of a pentagon?
  - Could the pentagon represent more than one value? Explain your thinking.

#### 8.3.1 Warm-Up: Balancing Hangers

The picture represents two balanced hangers. The hangers are balanced because the weight on each side is the same.



- In both hanger diagrams, a triangle weighs 3 grams and a circle weighs 6 grams.
  - Find the weight of a square in Hanger A.  
 $\square = 3.75$  grams
  - Write an equation to represent each hanger, where  $s$  represents the weight of each square and  $p$  represents the weight of each pentagon, in grams.

| Hanger | Equation            |
|--------|---------------------|
| A      | $21 + s = 5s + 6$   |
| B      | $12 + 2p = 2p + 12$ |



8.3.2 Exploration: Equations with Variables on Both Sides

Similar to hanger diagrams, algebra tiles can be used to model equations. As seen in the Warm-Up, it is possible to have linear equations with variables on both sides of the equal sign.

Work with your partner to use algebra tiles to explore how to solve such equations.

- 1. Use algebra tiles to solve the equation  $4x = 2x + 12$ .
  - a. Complete the table to describe the process for solving each equation.

| Algebra Tile Model | Description                                                   | Algebraic Equation                             |
|--------------------|---------------------------------------------------------------|------------------------------------------------|
|                    | Given equation                                                | $4x = 2x + 12$                                 |
|                    | Add $-2x$ to both sides of the equation to create zero pairs. | $4x = 2x + 12$<br>$+2x \quad +2x$<br>$2x = 12$ |
|                    | Divide both sides of the equation into <u>2</u> groups.       | $\frac{2x}{2} = \frac{12}{2}$                  |

8.3.2 EXPLORATION: EQUATIONS WITH VARIABLES ON BOTH SIDES

TEACHER MOVES

Students could use the images in the lesson, virtual algebra tiles, or physical algebra tiles to engage with the tasks in this component.

Encourage students to verbally describe each step of solving the equations with their partner.

TEACHER MOVES

8.3.2 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new equation.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group Guided Instruction. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

### 8.3.2 EXPLORATION: EQUATIONS WITH VARIABLES ON BOTH SIDES (CONTINUED)

#### TEACHER MOVES

##### Compare and Connect

Encourage students to explain how the algebra tiles specifically represent each part of the equation. Also allow students to use the tiles to explore different methods they could use to solve the equation.

- b. What is the solution to  $4x = 2x + 12$  based on the work done in the table in part a?  
 $x = 6$
2. Use algebra tiles to solve the equation  $8x + 15 = 4x - 1$ .
- a. Complete the table to describe the process for solving each equation.

| Algebra Tile Model | Description                                                   | Algebraic Equation                                            |
|--------------------|---------------------------------------------------------------|---------------------------------------------------------------|
|                    | Given equation                                                | $8x + 15 = 4x - 1$                                            |
|                    | Add $-4x$ to both sides of the equation to create zero pairs. | $8x + 15 = 4x - 1$<br>$\quad -4x \quad -4x$<br>$4x + 15 = -1$ |
|                    | Add $-15$ to both sides of the equation to create zero pairs. | $4x + 15 = -1$<br>$\quad -15 \quad -15$<br>$4x = -16$         |
|                    | Divide both sides of the equation into $\frac{1}{4}$ groups   | $\frac{4x}{4} = \frac{-16}{4}$                                |

- b. What is the solution to  $8x + 15 = 4x - 1$ , based on the work done in the table in part a?  
 $x = -4$

3. Use algebra tiles to solve the equation  
 $6 - 3x = 8 + (-4x) + 2$ .  
 a. Use the process of solving the equation with algebra tiles to complete the algebraic representation of the process.

| Algebra Tile Model | Algebraic Representation                                       |
|--------------------|----------------------------------------------------------------|
|                    | $6 - 3x = 8 + (-4x) + 2$                                       |
|                    | $6 - 3x = 8 + (-4x) + 2$<br>$+3x \quad +3x$<br>$6 = 8 - x + 2$ |
|                    | $6 = 8 - x + 2$<br>$6 = 10 - x$                                |
|                    | $6 = 10 - x$<br>$-6 \quad -6$<br>$-4 = -x$                     |
|                    | $-4 = -x$                                                      |

## 8.3.2 EXPLORATION: EQUATIONS WITH VARIABLES ON BOTH SIDES (CONTINUED)

### TEACHER MOVES

Avoid using the word “canceling,” as it is not mathematically accurate language. Sometimes “cancel” is used to mean a term is eliminated (makes 0), while other times it is used to mean a coefficient is made to be 1 (not eliminated). This can create confusion for students about what it means to “cancel.” It is best to instead use the language of “eliminating terms” using “zero pairs” and isolating the variable with a coefficient of 1.

# 8.3.2 EXPLORATION: EQUATIONS WITH VARIABLES ON BOTH SIDES (CONTINUED)

## TEACHER MOVES

After students have the opportunity to discuss and draft a list with their partners, facilitate a whole-group discussion to share some of the similarities and differences students have come up with.

- b. Marcus said both sides of the last equation in the table should be either multiplied or divided by  $-1$ . Explain whether you agree with Marcus' recommendation.  
*Answers will vary. I agree because the  $x$ -term has a coefficient of  $-1$  that must be eliminated to isolate the variable.*
- c. Based on your work in Parts A and B, what is the solution to  $6 - 3x = 8 + (-4x) + 2$ ?  
 $x = 4$

4. Discuss with your partner the similarities and differences between solving equations with variables on only one side and solving equations with variables on both sides. Record similarities and differences discussed.

| Similarities                                                                                  | Differences                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal is always to isolate the variable.                                                       | There are typically more steps.                                                                                                                                                     |
| Whatever is done to one side of the equation is also done to the other to maintain balance.   | To isolate the variable, it's helpful to isolate all constant terms on one side of the equation and all variable terms on the other by adding/subtracting quantities to both sides. |
| Terms can be added or subtracted from both sides to form zero pairs.                          |                                                                                                                                                                                     |
| Multiplication or division can be used to isolate a variable with a coefficient other than 1. |                                                                                                                                                                                     |



### 8.3.3 Guided Instruction: Solving Equations with Variables on Both Sides

1. Solve the equation  $3y - 4 = 6 - 2y$  by following the steps described in the table.

| Description                                                                 | Algebraic Representation |
|-----------------------------------------------------------------------------|--------------------------|
| Begin with the given equation.                                              | $3y - 4 = 6 - 2y$        |
| Eliminate the constant term from one side of the equation using zero pairs. | $3y = 10 - 2y$           |
| Eliminate the variable term from one side of the equation using zero pairs. | $5y = 10$                |
| Isolate the variable.                                                       | $y = 2$                  |

For some multi-step linear equations, it may be helpful to apply the distributive property or combine like terms on one side of the equation first.

2. Solve the equation  $4(m + 7.2) = -2m - 8.1$  by following the steps described in the table.

| Description                                                | Algebraic Representation |
|------------------------------------------------------------|--------------------------|
| Begin with the given equation.                             | $4(m + 7.2) = -2m - 8.1$ |
| Apply the distributive property.                           | $4m + 28.8 = -2m - 8.1$  |
| Eliminate the variable term from one side of the equation. | $6m + 28.8 = -8.1$       |
| Eliminate the constant term from one side of the equation. | $6m = -36.9$             |
| Isolate the variable.                                      | $m = -6.15$              |

3. Solve the equation  $-12p + 3p - 11 = 14 - 3p - 7$ .  
 $p = -3$

## 8.3.3 GUIDED INSTRUCTION: SOLVING EQUATIONS WITH VARIABLES ON BOTH SIDES

### DIFFERENTIATE INSTRUCTION

If students seem ready after guiding them through the first two problems, consider releasing them to student pairs to solve the third equation. Circulate to observe different ways in which students solve the equation and identify methods to share as well as misconceptions to address.

Consider allowing students to use whiteboards to work out problems, which will make it easier to highlight student methods when debriefing.



## 8.3.4 YOUR TURN!

## DIFFERENTIATE INSTRUCTION

If students need additional support, allow them to continue using algebra tiles to model and solve the equations.



## 8.3.4 Your Turn!

Solve each equation. Verify your solution using substitution.

1.  $6 - x = 5x + 30$   
 $x = -4$

$$6 - (-4) = 5(-4) + 30$$

$$10 = -20 + 30$$

$$10 = 10$$

2.  $9(p - 4) = 3(2p + 6)$   
 $p = 18$

$$9(18 - 4) = 3(2(18) + 6)$$

$$9(14) = 3(42)$$

$$126 = 126$$



### 8.3.5 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate how you feel about today's learning target.

I can solve equations with variables on both sides.

←—————→

I need help.      I'm getting there.      I understand and  
I can't solve      have evidence.  
equations with  
variables on both  
sides.

2. Explain your placement of "X" on the scale.  
*Answers will vary.*

## 8.3.5 WRAP-UP: REFLECTION

### DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

### QUESTIONING

What would reflection look like for you as a teacher? Consider your own scale for strengths and gaps for students through the lesson and how best to encourage students' confidence in solving multi-step equations in one variable.